Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Constructing Parallel Lines

Two lines are **parallel** when they stay the same distance apart all along their length — they never meet, no matter how far you extend them. In construction we draw parallel lines using a **set square** and a **ruler**.

The trick is that the set square has one straight edge, and the ruler gives you a straight, fixed line to slide the set square along. When you slide the set square along a ruler, the edge of the set square that you draw along keeps the same direction the whole time — so every line you draw with it stays parallel to every other line you draw with it.

Here is the rule to remember: **the ruler does not move, and the set square slides against it.**

You will need: a sharp pencil, a ruler, and a set square from your mathematical set.

Example — construct a line through point P parallel to line AB

Step 1. Draw the line AB with your ruler. Mark a point P anywhere above it. This is the point through which your new parallel line must pass.

Step 2. Place the set square so that one of its straight edges lies exactly on the line AB.

Step 3. Hold the set square firmly in that position. Place the ruler against one of the *other* edges of the set square, so the ruler is snug against it and the set square can still slide.

Step 4. Press the ruler down hard so it does not move. Now slide the set square along the ruler — up toward point P. As you slide, the edge that was on AB stays pointing in the same direction.

Step 5. Slide the set square until that edge reaches point P. Draw a line along the edge of the set square through P.

That new line is parallel to AB, because it was drawn with the very same edge of the set square that lay on AB, so both lines have the same direction.

Practice

1. Draw a horizontal line XY about 6 cm long. Using your set square and ruler, construct a line through a point Q that you mark above the line, so that the new line is parallel to XY. Explain in one sentence why your new line is parallel to XY.
2. Draw a slanted line AB. Construct a line through a point P below it that is parallel to AB. Which piece of your mathematical set must you hold still the whole time, and which piece slides?

Section 2: Drawing Perpendicular Lines

Two lines are perpendicular when they meet at a right angle — that is, at an angle of 90°. You can think of the corner of a book or a sheet of paper: the two edges that meet there are perpendicular to each other.

The best tool for drawing a perpendicular line is the set square, because it has one right-angled corner. You will use it together with a ruler, so that your drawn line stays straight.

Here is the plan for drawing a perpendicular line at a chosen point on a straight line.

1. Draw the given line. Use your ruler to draw a straight line. Mark the point where you want the perpendicular to meet it.
2. Line up the set square. Put one edge of the set square along your drawn line, with the right-angle corner sitting on your marked point. The set square's other right-angle edge should stand up from the line.
3. Draw the perpendicular. Hold the set square firmly and draw along the standing edge. The line you draw rises from your point and makes a right angle with the first line.

Because the set square's corner is exactly 90°, the new line is perpendicular to the old one. Check your work: the two lines should form four right angles where they cross, just like the corner of a page.

Worked example

Construct a perpendicular to the line below at the point marked P. You will do this on paper with your ruler and set square.

P

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Solution

1. Place the ruler along the given line so one long edge lies exactly on it.
2. Put the set square on top of the ruler so that one of its right-angle edges lies against the ruler's edge, and the right-angle corner touches the ruler where point P is.
3. Keeping the set square pressed against the ruler, draw along the set square's upright edge, starting at P and going upwards.
4. The upright line and the original line meet at P, and the corner between them is a right angle.

Your drawing should now look like this, with a little square drawn in the corner to mark the right angle:

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  P
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The line you drew is perpendicular to the original line.

Practice

1. Draw a straight horizontal line about 10 cm long. Mark a point in the middle of it. Using your ruler and set square, construct a perpendicular line at that point. What angle do the two lines make where they meet?
2. Which tool do you use to draw a perpendicular line, and why is it suitable?

Section 3: Bisecting a Line Segment

To bisect a line segment means to divide it into two equal parts. The point that cuts the segment into two equal halves is its midpoint. You can find it exactly with a pair of compasses, without measuring.

Here is the method for a segment from point A to point B:

1. Draw the line segment AB.
2. Open the compass to a distance more than half the length of AB. (If you open it less than half, the arcs below will not meet.)
3. Place the compass point on A. Draw an arc that swings over the line and one that swings under the line.
4. Keeping the compass opening exactly the same, place the compass point on B. Draw arcs that cross the first two — one above the line and one below it.
5. The two crossing points — one above the line and one below it — mark where the arcs meet.
6. Draw a straight line through those two crossing points. Where this line meets AB is its midpoint, M. The segment has been bisected: AM and MB are equal.

The secret of the method is that you never change the compass opening between steps 3 and 4. That is what makes the two crossing points line up exactly through the midpoint.

Worked example — bisecting PQ

Follow each step on your own paper with your compass set.

• Draw a segment and label its ends P and Q.
• Open the compass to a little more than half the length of PQ.
• Put the point on P and swing an arc above the segment and an arc below it.

• Without changing the compass opening, put the point on Q and swing arcs that cross the first two — one above, one below.
• You now have two crossing points, one on each side of the segment. Draw the straight line through both crossings.
• The line you drew cuts PQ at its midpoint. Label it M. Measure PM and MQ with a ruler — they are equal.

Practice

1. Draw a segment AB of any length. Bisect it with a pair of compasses, then label the midpoint M. State what you can say about the lengths AM and MB.
2. To bisect a segment CD, you open your compass to a distance. Must this distance be less than, equal to, or more than half the length of CD? Explain why.
3. While bisecting a segment, after drawing arcs from the first end you change the compass opening before drawing arcs from the second end. Will the line through the two crossing points still cut the segment exactly in half? Give a reason.

Section 4: Constructing a Right Angle (90°)

A right angle measures exactly 90°, and you can build one wherever you like on a straight line using only your mathematical set. Here you will construct the 90° angle at a chosen point on a line, and you will use compass arcs to do it.

Every angle you draw needs two arms meeting at a point (the vertex). To construct 90° you will raise a new arm straight up from a point that already sits on your line, so that the new arm and the original line make a perfect right angle.

The whole trick is to work on both sides of your point with the compass, then join across. The steps you will follow are:

1. Draw a straight line and mark a point P on it — P is where the right angle will sit.
2. Place your compass point on P and open it to a convenient radius. Draw one arc cutting the line on the left of P and one arc cutting the line on the right of P. Call these two cuts M and N. (M and N are now equal distances from P.)
3. Open your compass a little wider than before. Place the point on M and draw an arc above the line.
4. Keep the same radius, move the compass point to N, and draw another arc above the line so that it crosses the first arc. Call the crossing point C.
5. Use your ruler to draw a straight line from P through C.

The line PC is now at right angles to your original line, so the angle where P, the line and PC meet is exactly 90°. Because you used the same compass opening for both arcs in step 3 and 4, C stands exactly above P, and that is what guarantees the right angle.

Let us construct the 90° angle together on a fresh line, so you can see the full construction before you try one alone.

Take a ruler and draw a straight horizontal line across your page. Mark a point P about halfway along it.

Now set your compass to a comfortable radius (about 3 cm) and place the metal point on P. Swing the compass so the pencil draws one small arc that cuts the line to the left of P, and then swing it again to cut the line to the right of P. Label the left cut M and the right cut N. Because you did not change the compass opening, PM and PN are equal.

Next, open your compass a little wider (say 4 cm). Put the point on M and draw a fair-sized arc that rises above the line. Without changing the opening, move the point to N and draw another arc above the line. The two arcs should cross somewhere above P — call that crossing point C.

Finally, lay your ruler so its edge passes through P and C, and draw the straight line PC, letting it run above and below P.

The two lines now cross at P, and the angle between the right-hand side of your original line and the arm PC is 90°. You have constructed a right angle.

Practice

1. On a fresh line, construct a 90° angle at a point P that lies on the line. When you are finished, explain how you know the angle you drew is exactly 90°.
2. When constructing a 90° angle at a point P on a line, you first cut the line at two points M and N using one fixed compass opening. Why must you keep the compass at the same opening for these two cuts, and what does that guarantee about M and N?

Section 5: Constructing a 60° Angle with Compasses

A 60° angle is one of the special angles you can draw with just a ruler and a pair of compasses — no protractor needed. The whole trick is to build an equilateral triangle: a triangle whose three sides are all equal. In an equilateral triangle, every angle is 60°. So if you draw an equilateral triangle and keep one of its angles, that angle is your 60°.

Here is the construction, step by step. You will need a sharp pencil, a ruler, and your compasses (part of your mathematical set).

Step 1. Draw a straight line. Choose a point near one end of it and call it A. This point will be the corner where your 60° angle sits.

Step 2. Set your compasses to any radius that fits on your page — the size does not matter, as long as it stays the same throughout. Place the compass point on A and draw an arc that crosses the line. Call the point where the arc meets the line B.

Step 3. Without changing the compass setting, place the compass point on B and draw another arc. Make it cross the first arc. Call the point where the two arcs meet C.

Step 4. Use your ruler to draw a straight line from A to C.

The angle between the original line and the line AC — that is, angle BAC — is 60°.

Why does this work? In Step 2, the distance from A to B is your chosen radius, so AB is one radius long. In Step 3, the second arc is drawn with the same radius from B, so BC is also one radius long. And in Step 4, the line AC reaches the point C that lies on the first arc from A, so AC is one radius long too. That means AB = BC = AC: the triangle is equilateral, and every angle in it, including angle BAC, is exactly 60°.

Worked example — construct a 60° angle at point A.

1. Draw a horizontal line on your page and mark a point A near the middle.
2. Set your compasses to a radius of about 4 cm (any size works). Put the point on A and draw an arc that crosses the line to the right of A. Mark where it lands as B.
3. Keep the compasses exactly as they are. Put the point on B and draw an arc. Make sure it crosses the first arc above the line. Mark the meeting point as C.
4. Draw a straight line from A through C, extending it a little.

You now have angle BAC between the original line and line AC. Measure it with your protractor to check — it should read 60°. The two arcs must use the same radius, so never change your compass setting between Step 2 and Step 3.

Practice

1. Construct a 60° angle following the compass-arc method you have just learned. Use your protractor afterwards to check the size of your angle. What should your angle measure?
2. In the construction of a 60° angle, the three points A, B and C form a triangle whose three sides are all equal. What is the name of this kind of triangle, and what is the size of each of its angles?
3. Suppose you set your compasses to a radius of 5 cm and complete the full construction of a 60° angle. How long is each side of the triangle formed by points A, B and C?

Check your work

Attempt every question before you read this page.

Constructing Parallel Lines

1. Your new line through Q is parallel to XY because it was drawn with the same edge of the set square that lay on XY, so it keeps the same direction all along — the two lines never meet, no matter how far they are extended.

   Place the set square so one edge lies exactly on XY, put the ruler against another edge and hold it still, slide the set square up to Q, then draw along the set square's edge through Q. Since the set square only slid (it did not turn), the drawing edge kept its direction, so the line through Q has the same direction as XY — that is what makes them parallel.
2. The ruler must be held still the whole time, and the set square slides along it. The edge of the set square that was on AB keeps its direction as it slides, so the line drawn through P is parallel to AB.

Drawing Perpendicular Lines

1. 90° (a right angle). The lines meet at four right angles.

   After drawing the horizontal line, place the set square so one right-angle edge lies along the line and the right-angle corner sits on the marked point. Draw along the upright edge. The two lines form a right angle at the meeting point, which measures 90°.
2. The set square (with the ruler to keep the line straight), because it has a right-angled corner that is exactly 90°.

   A set square has one corner that is exactly a right angle. Lining that corner up on the given line guarantees the new line is perpendicular.

Bisecting a Line Segment

1. AM and MB are equal.

   Draw AB, then with the compass opened more than half of AB, draw arcs above and below from A, then the same arcs from B. Draw the line through the two crossing points; its meeting with AB is M. Because the construction keeps the compass opening the same, M lies exactly at the middle, so AM = MB.
2. More than half.

   If the opening were less than half of CD, the arcs drawn from C and D would not reach across to meet one another on both sides of the segment. Opening it more than half guarantees the arcs from the two ends cross, giving the two crossing points the bisecting line passes through.
3. No — the bisector must be drawn with the same compass opening at both ends.

   The whole construction works because both pairs of arcs are swung at the same distance from the two ends, so the crossing points sit equally far from A and B and therefore line up through the midpoint. If the opening changes, the crossing points shift off that central line, so the drawn line no longer passes through the midpoint.

Constructing a Right Angle (90°)

1. The angle at P between the original line and the line drawn from P through C is exactly 90° because C was located by two equal compass arcs swung from M and N, which sit at equal distances on either side of P. That places C directly above P, so the new arm PC is perpendicular to the original line, making the angle a right angle.
   a. Draw a straight line and mark point P on it. 2. Set the compass to a fixed radius, place the point on P, and cut the line at M (left) and N (right). 3. Open the compass wider; from M draw an arc above the line, then from N (same opening) draw an arc crossing it at C. 4. Draw the line from P through C. Since M and N are equidistant from P and the two arcs used the same radius, C lies directly above P, so PC is perpendicular to the line and the angle is 90°.
2. Keeping the compass at the same opening for the two cuts guarantees that PM and PN are equal — that is, M and N are the same distance from P on opposite sides. This balance is what places the crossing point C directly above P, so the line drawn from P through C stands perpendicular to the original line and the angle is exactly 90°.

A compass swung at a fixed opening draws points that are all the same distance from the centre point. So cutting at M with opening r gives PM = r, and cutting at N with the same opening r gives PN = r. Equal distances on either side of P balance the construction so the arc crossing C lands straight above P.

Constructing a 60° Angle with Compasses

1. 60°

   Draw a line, mark a point A, draw an arc from A to meet the line at B, draw an equal arc from B to meet the first arc at C, then join A to C. The angle between the original line and line AC is 60°, because AB, BC and AC are all equal radii, forming an equilateral triangle.
2. It is an equilateral triangle, and each of its angles is 60°.
3. Each side is 5 cm.

   The radius you chose is used for every arc, so AB is one radius (5 cm), BC is one radius (5 cm), and AC is one radius (5 cm). All three sides are equal, so each is 5 cm.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-f50f19ef	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____